

Tilen Žel

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📍 Location: Ljubljana, Slovenia

EDUCATION

University of Ljubljana

Master's Degree (Physics)

Faculty of Mathematics and Physics

SKILLS

Languages and platforms:

C/C++ , Python , Matlab , Julia , CUDA 

Other:

Microsoft Office, LaTeX/Typst, Medical Physics, Photonics

WORK EXPERIENCE

Fotona

Ljubljana

R&D Engineer

Sep 2021 - Present

- Made major improvements in area of numerical simulations, which has led to significant benefits for research and regulatory approval of various devices.
- Participated in research projects involving laser technology, laser treatments, and laser-tissue interaction

Faculty of Mechanical Engineering (FS)

Ljubljana

Research Aide

Oct 2018 - July 2021

- Assisted in developing a simulation of capillary tubes, which are a crucial component of various cooling devices, such as air conditioners and refrigerators.
- Analyzed the condensation of moist air in water condensers for different geometries and flow regimes.

Jozef Stefan Institute (IJS)

Ljubljana

Research assistant (Master's degree)

Dec 2019 - June 2021

- Analyzed spectroscopic measurements and optically scattering materials by using and modifying the Monte Carlo method in the CUDA platform and the C programming language.

PROJECTS

Magnetic simulation for arbitrary coils (Fotona): *Matlab, C, CUDA*

Modified a Matlab toolbox for calculation of magnetic field, improved it to enable calculation of induced current in tissue, and accelerated it by enabling its computation on graphic card using C and CUDA.

Simulation of capillary tubes (FS): *Matlab, C++, wxWidgets*

My work consisted of minor assistance in the creation of the original code in Matlab and re-implementation of the entire project to the C++ programming language, along with creation of a GUI.

Simulation of radio-frequency heating (Fotona): *Julia*

Created a simulation that determines 3D heat distribution in tissue after RF treatment for arbitrary (Dirichlet or Neumann) boundary conditions.

PUBLICATIONS

-  M. Lukač, J. Košir, T. Žel, M. Kažič, D. Šavli and M. Jezeršek, "Influence of tissue desiccation on critical temperature for thermal damage during Er:YAG laser skin treatments", *Lasers in Surgery and Medicine*, No. 56(1), pp. 107-118, Nov. 2023, doi: <https://doi.org/10.1002/lsm.23739>
-  B. Majaron and T. Žel, "Optical properties of Spectralon assessed by replication of literature data in Monte Carlo simulations", *Proceedings Volume 12147, Tissue Optics and Photonics II; 1214704*, Strasbourg, France, May 2022, doi: <https://doi.org/10.1117/12.2620704>
-  P. Poredoš *et al.*, "Condensation of water vapor from humid air inside vertical channels formed by flat plates", *iScience* 25(1), pp. 1--26, Jan. 2022, doi: <https://doi.org/10.1016/j.isci.2021.103565>